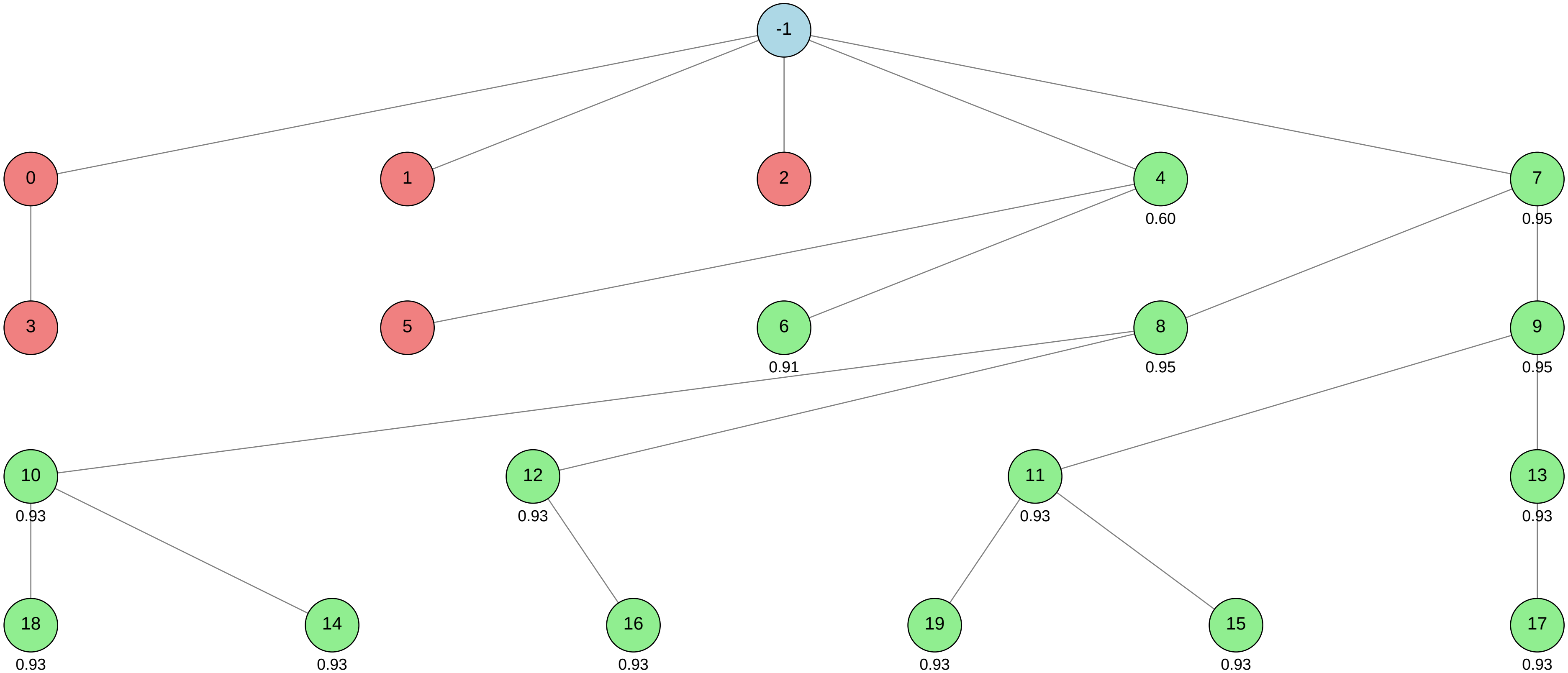
Node Structure Visualization

Click on any node in the tree to jump to its detailed information.

Total Nodes: 21 | Best Validation Score: 0.9458 (Node 7)

Node Tree:

Table of Contents:



Status	Color
Success	Green
Failure	Red
Neutral	Blue

Node Details:

Node -1 (root)

Property	Value
uct_value	0.6005
ctime	2026-05-21 16:14:46
debug_attempts	0
depth	0
execution_time	0.0
expected_child_count	0
failure_visits	5
id	-1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
num_children	5
prev_tutorial_prompt	None
stage	root
time_step	-1
tool_used	
tools_available	[]
unvalidated_visits	0
validated_reward	13.672800000000004
validated_visits	15
validation_score	None
visits	20
parent_id	None
child_ids	[0, 1, 2, 4, 7]

Node 0 (evolve)

Property	Value
uct_value	1.7306
ctime	2026-05-21 16:15:45
debug_attempts	1
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.394748179489867
failure_penalty	-0.0
failure_visits	2
id	0
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	None
stage	evolve
time_step	0
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	2
parent_id	-1
child_ids	[3]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 52-addition_3/node_0/conda_env
Resolved 235 packages in 1.13s
Uninstalled 1 package in 1.85s
Installed 233 packages in 1m 48s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260521
+ autogluon-common==1.5.1b20260521
+ autogluon-core==1.5.1b20260521
+ autogluon-features==1.5.1b20260521
+ autogluon-multimodal==1.5.1b20260521
+ autogluon-tabular==1.5.1b20260521
+ autogluon-timeseries==1.5.1b20260521
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.12
+ botocore==1.43.12
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0

+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.0
+ cloudpathlib==0.24.0
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7

+ idna==3.15
... [truncated, 13550 chars total]

Error Analysis:

ERROR_SUMMARY: The process was forcibly terminated after exceeding the 1-hour (3600 seconds) wall-clock time limit. Although model training was progressing and validation AUROC scores were being logged, the script did not finish all required steps, including final output validation and saving of prediction results. As a result, there is no guarantee that the output file was written or that all output checks passed.

SUGGESTED_FIX:
Reduce the total runtime by limiting the number of training epochs (e.g., set `"optim.max_epochs"` to a smaller value such as 2–4), lowering model complexity (e.g., use a lighter backbone or smaller batch size), or setting a strict `time_limit` parameter in `predictor.fit()` (e.g., `time_limit=3400`). Ensure that the script always saves predictions and performs output validation as soon as possible after training, even if training is interrupted or time-constrained.

Node 1 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-21 17:20:38
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2486433288776608
failure_penalty	-0.0
failure_visits	1
id	1
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	1
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt` does not contain any dependencies
Using Python 3.11.15 environment at: 52-addition_3/node_1/conda_env
Resolved 79 packages in 357ms
Uninstalled 1 package in 1.84s
Installed 78 packages in 36.34s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.0
+ idna==3.15
+ jinja2==3.1.6

+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocess==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparse==12.6.3.3
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.13.4
+ pydantic-core==2.46.4
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 6029 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the code attempts to load pretrained weights for the model 'efficientnetv2_s' using timm, but these weights are not available in the current timm installation. As a result, timm raises a RuntimeError instructing to use `pretrained=False` for random initialization or to choose a model for which pretrained weights exist.

SUGGESTED_FIX:
Modify the model creation line in your code to set `pretrained=False` when calling `timm.create_model('efficientnetv2_s', ...)`, or select a different model architecture that has available pretrained weights in your timm version. After making this change, rerun your script to proceed with random initialization or with a supported pretrained model.

Node 2 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-21 17:26:03
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2131672923786587
failure_penalty	-0.0
failure_visits	1
id	2
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	2
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 52-addition_3/node_2/conda_env
Resolved 239 packages in 2.69s
Uninstalled 1 package in 1.61s
Installed 237 packages in 1m 26s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260521
+ autogluon-common==1.5.1b20260521
+ autogluon-core==1.5.1b20260521
+ autogluon-features==1.5.1b20260521
+ autogluon-multimodal==1.5.1b20260521
+ autogluon-tabular==1.5.1b20260521
+ autogluon-timeseries==1.5.1b20260521
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.12
+ botocore==1.43.12
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0

+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.0
+ cloudpathlib==0.24.0
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ daal==2025.9.0
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2

+ hyperopt==0.... [truncated, 170878 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the test set does not contain the columns ['skin_tone', 'alternative_skin_tone', 'expert_opinion'], but the code tries to select these columns from the test DataFrame (`test[feature_cols]`). This results in a KeyError, as only 'Unnamed: 0' and 'image_name' are present in test.csv.

SUGGESTED_FIX:
Update the code to avoid referencing tabular features that do not exist in the test set. Instead, use only the columns available in test.csv (e.g., 'image_name') and ensure that the model is trained and predicts using features that are present in both train and test sets. For image-based tasks, switch to a multimodal or image-only approach (e.g., AutoGluon Multimodal), where the image file paths are used as input features, and do not expect tabular features in the test set.

Node 3 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-21 17:34:59
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
failure_visits	1
id	3
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	debug
time_step	3
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	0
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 52-addition_3/node_3/conda_env
Resolved 235 packages in 1.09s
Uninstalled 1 package in 1.71s
Installed 233 packages in 1m 33s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260521
+ autogluon-common==1.5.1b20260521
+ autogluon-core==1.5.1b20260521
+ autogluon-features==1.5.1b20260521
+ autogluon-multimodal==1.5.1b20260521
+ autogluon-tabular==1.5.1b20260521
+ autogluon-timeseries==1.5.1b20260521
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.12
+ botocore==1.43.12
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.0
+ cloudpathlib==0.24.0

```
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.15
... [truncated, 18398 chars total]
```

Error Analysis:

ERROR_SUMMARY: The root cause of the error is that PyTorch's distributed training backend (torch.distributed) failed to initialize because the required TCP port (20621) was already in use (EADDRINUSE). This prevented the distributed process group from starting, which is necessary when using multiple GPUs (env.num_gpus=2). As a result, model training could not proceed, and subsequent errors (such as _pickle.UnpicklingError and OSError: Device or resource busy) occurred as cascading failures from the initial distributed setup problem.

SUGGESTED_FIX:
Set "env.num_gpus" to 1 in your AutoGluon hyperparameters to disable distributed training, or explicitly specify a different, unused port for distributed initialization if multi-GPU is required. Before rerunning, ensure no other processes are occupying the target port, or restart the environment to clear any stale distributed training processes. If you do not require multi-GPU training, using a single GPU will avoid this class of errors entirely.

Node 4 (debug)

Property	Value
uct_value	1.7104
avg_raw_score	0.7546999999999999
ctime	2026-05-21 17:44:18
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.29739130434782596
exploration	1.050633694240302
failure_penalty	-0.0
failure_visits	1
id	4
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.44608695652173896
num_children	2
prev_tutorial_prompt	None
stage	debug
time_step	4
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.29739130434782596
validated_reward	1.5093999999999999

validated_visits	2
validated_weight	0.6666666666666666
validation_score	0.6008
visits	3
parent_id	-1
child_ids	[5, 6]

Node 5 (evolve)

Property	Value
uct_value	1.4821
ctime	2026-05-21 18:28:25
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	5
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure yo # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	evolve
time_step	5
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	4
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt` does not contain any dependencies
Using Python 3.11.15 environment at: 52-addition_3/node_5/conda_env
Resolved 79 packages in 2.00s
Uninstalled 1 package in 3.56s
Installed 78 packages in 4m 08s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.0
+ idna==3.15

+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocess==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparse==12.6.3.3
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.13.4
+ pydantic-core==2.46.4
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ set... [truncated, 15868 chars total]

Error Analysis:

ERROR_SUMMARY: The main error is that the process was killed after exceeding the 1-hour (3600 seconds) wall-clock time limit, resulting in incomplete training (only 3 out of 4 epochs finished) and no evidence of completed test predictions or output validation. Additionally, there are repeated OSError: [Errno 16] Device or resource busy errors when attempting to delete temporary files (likely NFS-related), and warnings about excessive DataLoader worker processes (16 used, but only 2 recommended for the system).

SUGGESTED_FIX:
Reduce the number of DataLoader workers to 2 or fewer to avoid resource contention and potential slowness/freezing. Lower the batch size and/or the number of training epochs to ensure the script completes within the 1-hour time limit. Monitor and, if possible, explicitly close or release file handles before deletion to mitigate the "device or resource busy" errors with NFS temporary files.

Node 6 (evolve)

Property	Value
uct_value	2.3743
ctime	2026-05-21 19:32:17
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	0
id	6
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	evolve
time_step	6
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.9086
validated_visits	1

validation_score	0.9086
visits	1
parent_id	4
child_ids	[]

Node 7 (debug)

Property	Value
uct_value	1.6493
avg_raw_score	0.9359000000000002
ctime	2026-05-21 20:33:38
debug_attempts	2
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.9713043478260875
exploration	0.700422462826868
failure_penalty	-0.0
failure_visits	0
id	7
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9713043478260875
num_children	2
prev_tutorial_prompt	None
stage	debug
time_step	7
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.9713043478260875
validated_reward	12.163400000000003

validated_visits	13
validated_weight	1.0
validation_score	0.9458
visits	13
parent_id	-1
child_ids	[8, 9]

Node 8 (evolve)

Property	Value
uct_value	1.8926
avg_raw_score	0.9348
ctime	2026-05-21 21:00:07
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.9681159420289854
exploration	0.9099736640840701
failure_penalty	-0.0
failure_visits	0
id	8
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9681159420289854
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	8
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9681159420289854
validated_reward	5.6088
validated_visits	6
validated_weight	1.0
validation_score	0.9458
visits	6
parent_id	7
child_ids	[10, 12]

Node 9 (evolve)

Property	Value
uct_value	1.8926
avg_raw_score	0.93524
ctime	2026-05-21 21:28:29
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.969391304347826
exploration	0.9968262051089475
failure_penalty	-0.0
failure_visits	0
id	9
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.969391304347826
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	9
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9693913043478260
validated_reward	5.6088
validated_visits	6
validated_weight	1.0
validation_score	0.9458
visits	6
parent_id	7
child_ids	[11, 13]

Node 10 (evolve)

Property	Value
uct_value	2.0545
avg_raw_score	0.9326
ctime	2026-05-21 22:10:21
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9617391304347827
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	0
id	10
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9617391304347827
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	10
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9617391304347827
validated_reward	2.7978
validated_visits	3
validated_weight	1.0
validation_score	0.9326
visits	3
parent_id	8
child_ids	[18, 14]

Node 11 (evolve)

Property	Value
uct_value	2.0545
avg_raw_score	0.9326
ctime	2026-05-21 22:28:45
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9617391304347827
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	0
id	11
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9617391304347827
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	11
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9617391304347827
validated_reward	2.7978
validated_visits	3
validated_weight	1.0
validation_score	0.9326
visits	3
parent_id	9
child_ids	[19, 15]

Node 12 (evolve)

Property	Value
uct_value	2.3001
avg_raw_score	0.9326
ctime	2026-05-21 22:57:42
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9617391304347827
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	0
id	12
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9617391304347827
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	12
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9617391304347827
validated_reward	1.8652
validated_visits	2
validated_weight	1.0
validation_score	0.9326
visits	2
parent_id	8
child_ids	[16]

Node 13 (evolve)

Property	Value
uct_value	2.3001
avg_raw_score	0.9326
ctime	2026-05-21 23:23:38
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9617391304347827
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	0
id	13
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9617391304347827
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	13
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9617391304347827
validated_reward	1.8652
validated_visits	2
validated_weight	1.0
validation_score	0.9326
visits	2
parent_id	9
child_ids	[17]

Node 14 (evolve)

Property	Value
uct_value	2.4438
ctime	2026-05-21 23:45:21
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	14
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	14
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.9326
validated_visits	1
validation_score	0.9326
visits	1
parent_id	10
child_ids	[]

Node 15 (evolve)

Property	Value
uct_value	2.4438
ctime	2026-05-22 00:09:06
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	15
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	15
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.9326
validated_visits	1
validation_score	0.9326
visits	1
parent_id	11
child_ids	[]

Node 16 (evolve)

Property	Value
uct_value	2.1390
ctime	2026-05-22 00:54:34
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	16
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	16
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.9326
validated_visits	1
validation_score	0.9326
visits	1
parent_id	12
child_ids	[]

Node 17 (evolve)

Property	Value
uct_value	2.1390
ctime	2026-05-22 01:15:25
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	17
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	17
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.9326
validated_visits	1
validation_score	0.9326
visits	1
parent_id	13
child_ids	[]

Node 18 (evolve)

Property	Value
uct_value	2.4438
ctime	2026-05-22 01:36:01
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	18
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	18
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.9326
validated_visits	1
validation_score	0.9326
visits	1
parent_id	10
child_ids	[]

Node 19 (evolve)

Property	Value
uct_value	2.4438
ctime	2026-05-22 01:58:20
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	19
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	evolve
time_step	19
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.9326
validated_visits	1
validation_score	0.9326
visits	1
parent_id	11
child_ids	[]